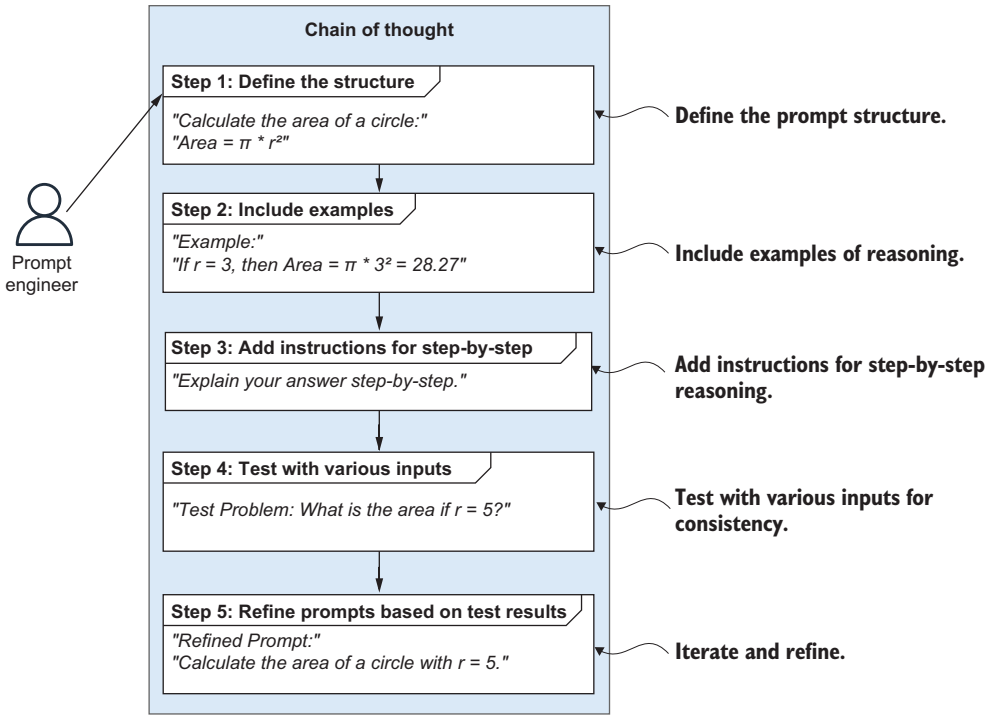


Figure 5.7 The key steps involved in creating chain-of-thought prompts that encourage LLMs to articulate their reasoning process step-by-step

2 *Provide examples.* Here we incorporate few-shot learning techniques, which include relevant examples that demonstrate how to approach similar problems. These examples serve as a guide for the model, helping it understand the expected reasoning process and the level of detail required in the final response. For example, "Example: If r = 3, then Area = $\pi * 3^2 = 28.27$."

   This example walks through the calculation of the area of a circle with a radius of 3 using the provided formula.

3 *Ensure step-by-step reasoning.* To ensure that the model provides a clear and logical explanation of its reasoning, it's important to include explicit instructions that prompt it to break down the problem into individual steps. For example, "Explain your answer step-by-step."

   By adding this directive, you encourage the model to explain its thought process in a structured manner.

4 *Test for consistency.* Before finalizing the prompt, it's important to verify it with various inputs to test that the model consistently follows the specified format and reasoning process. If you find any inconsistencies or errors in the model's responses,